

CROSS-VALIDATION

Cross-validation is a technique used to evaluate how well a machine learning model will generalize to unseen data. The main types of cross-validation are:

1. **Hold-Out Validation**
 - The dataset is split into two parts:
 - Training set (e.g., 80%)
 - Testing set (e.g., 20%)
 - The model is trained on the training set and evaluated on the test set.
 - **Advantage:** Fast and simple.
 - **Disadvantage:** Performance depends on the particular split.
2. **K-Fold Cross-Validation**
 - The dataset is divided into **K equal folds** (commonly $K = 5$ or 10).
 - The model is trained K times:
 - Each fold is used once as the validation set.
 - The remaining $K-1$ folds are used for training.
 - The final performance is the average of all K results.
 - **Advantage:** More reliable than hold-out validation.
3. **Stratified K-Fold Cross-Validation**
 - Similar to K-fold, but each fold maintains the same class distribution as the original dataset.
 - Commonly used for **classification problems with imbalanced classes**.
4. **Leave-One-Out Cross-Validation (LOOCV)**
 - A special case of K-fold where **$K = \text{number of samples}$** .
 - Each iteration uses one sample for testing and all remaining samples for training.
 - **Advantage:** Uses almost all data for training.
 - **Disadvantage:** Computationally expensive.
5. **Leave-P-Out Cross-Validation (LPOCV)**
 - Leaves **P samples** out for testing in each iteration.
 - Trains on the remaining samples.
 - More computationally intensive than LOOCV.
6. **Repeated K-Fold Cross-Validation**
 - K-fold cross-validation is repeated multiple times with different random splits.
 - The average performance over all repetitions is reported.
 - Improves the stability of performance estimates.
7. **Time Series Cross-Validation (Forward Chaining/Rolling Validation)**
 - Designed for **time-dependent data**.
 - Training is performed on earlier observations, and testing is done on later observations.
 - Prevents data leakage by preserving the chronological order.